

2.2.18

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Question

The unit vector normal to the plane
 $x + 2y + 3z - 6 = 0$
is:

Theoretical Solution

The normal vector to the plane is

$$\mathbf{n} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad (1)$$

The magnitude of $\mathbf{n}$ is

$$\|\mathbf{n}\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}. \quad (2)$$

Theoretical Solution

Therefore, the unit normal vectors are

$$\pm \frac{\mathbf{n}}{\|\mathbf{n}\|} = \pm \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad (3)$$

Main C Code

```
#include <stdio.h>
#include <math.h>

double magnitude(int x, int y, int z) {
    return sqrt(x*x + y*y + z*z);
}

void unitVector(int x, int y, int z) {
    double mag = magnitude(x, y, z);
    printf("Unit normal vector = (%.3f)i + (%.3f)j + (%.3f)k\n",
        x/mag, y/mag, z/mag);
}

int main() {
    int a = 1, b = 2, c = 3; // coefficients of x, y, z
    unitVector(a, b, c);
    return 0;
}
```

C Function

```
#include <math.h>

double magnitude(int x, int y, int z) {
    return sqrt(x*x + y*y + z*z);
}

void unitVector(int x, int y, int z, double *ux, double *uy,
    double *uz) {
    double mag = magnitude(x, y, z);
    *ux = x / mag;
    *uy = y / mag;
    *uz = z / mag;
}
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# === Load the C shared library ===
lib = ctypes.CDLL("./libunitvector.so")

# Define C function signature
lib.unitVector.argtypes = (ctypes.c_int, ctypes.c_int, ctypes.c_int,
                           ctypes.POINTER(ctypes.c_double),
                           ctypes.POINTER(ctypes.c_double),
                           ctypes.POINTER(ctypes.c_double))
lib.unitVector.restype = None
```

```
# Plane equation:  $ax + by + cz + d = 0$ 
a, b, c, d = 1, 2, 3, -6 # Example:  $x + 2y + 3z - 6 = 0$ 

# Output variables for unit vector
ux = ctypes.c_double()
uy = ctypes.c_double()
uz = ctypes.c_double()

# Call C function to get unit normal vector
lib.unitVector(a, b, c, ctypes.byref(ux), ctypes.byref(uy),
               ctypes.byref(uz))
u = np.array([ux.value, uy.value, uz.value])

# === Plotting ===
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
```



```
# Create mesh grid for the plane
xx, yy = np.meshgrid(range(-2, 5), range(-2, 5))
zz = -(a*xx + b*yy + d) / c # Solve for z from ax+by+cz+d=0

# Plot the plane
ax.plot_surface(xx, yy, zz, alpha=0.5, color='cyan')

# Pick a point on the plane (x=0,y=0 z=-d/c)
point = np.array([0, 0, -d/c])

# Plot the normal vector
ax.quiver(point[0], point[1], point[2],
          u[0], u[1], u[2],
          length=2, color='red', linewidth=2)
```

```
1 ax.set_xlabel("X")
2 ax.set_ylabel("Y")
3 ax.set_zlabel("Z")
4 ax.set_title("Plane and its Unit Normal Vector")
5 plt.savefig("/home/r-nikhil/ee1030-2025/ai25btech11025/matgeo
6 /2.2.18/figsplotc.png")
7 plt.show()
```

Plane and its Unit Normal Vector

